

AIC/AICc/BIC Formula in R for GLM

Asked 4 years ago Modified 4 years ago Viewed 8k times  Part of R Language Collective



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I'm trying to check that I understand how R calculates the statistic AIC, AICc (corrected AIC) and BIC for a `glm()` model object (so that I can perform the same calculations on `revoscaleR::rxGlm()` objects - particularly the AICc, which isn't available by default)



I had understood that these were defined as follows:



let p = number of model parameters



let n = number of data points

```
AIC = deviance + 2p
AICc = AIC + (2p^2 + 2p)/(n-p-1)
BIC = deviance + 2p.log(n)
```

So I tried to replicate these numbers and compare them to the corresponding R function calls. It didn't work:

```
library(AICcmodavg) # for the AICc() function

data(mtcars)

glm_a1 <- glm(mpg ~ cyl + disp + hp + drat + wt + qsec + vs + am + gear + carb,
             data = mtcars,
             family = gaussian(link = "identity"),
             trace = TRUE)

summary(glm_a1)

n <- nrow(glm_a1$data) # 32
p <- glm_a1$rank # 11
dev <- glm_a1$deviance # 147.49

my_AIC <- dev + 2 * p
my_AICc <- my_AIC + (2 * p^2 + 2 * p)/(n - p - 1)
my_BIC <- dev + 2 * p * log(n)

AIC(glm_a1) # 163.71
my_AIC # 169.49

AICc(glm_a1) # 180.13 (from AICcmodavg package)
my_AICc # 182.69

BIC(glm_a1) # 181.30
my_BIC # 223.74
```

By using `debug(AIC)` I can see that the calculation is different. It's based on 12 parameters (one extra for the estimated dispersion/scale parameter?). Also the log likelihood is obtained using `logLik()` which brings back a number -69.85 , which suggests to me that the model deviance would be $-2 \times -69.85 = 139.71$ (which it isn't).

Does anyone know what I've done wrong please? Thank you.

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2 Answers

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Function to calculate these quantities for an `rxGlm()` object consistent with treatment of `glm()` (adjusting for the "up to a constant" difference in deviance):

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```
wrc_information_criteria <- function(rx_glm) # an object created by rxGlm()
{
  # add 1 to parameter count for cases where the GLM scale parameter needs to be
  # estimated (notably Gamma/gaussian)

  extra_parameter_flag <- case_when(
    rx_glm$family$family == "gaussian" ~ 1,
    rx_glm$family$family == "Gamma" ~ 1,
    rx_glm$family$family == "poisson" ~ 0,
    rx_glm$family$family == "binomial" ~ 0,
    TRUE ~ 999999999
  )

  n <- rx_glm$nValidObs
  p <- rx_glm$rank + extra_parameter_flag
  dev <- rx_glm$deviance

  cat("\n")
  cat("n      :", n, "\n")
  cat("p      :", p, "\n")
  cat("deviance:", dev, "\n")

  AIC <- dev + 2 * p
  AICc <- AIC + (2 * p^2 + 2 * p)/(n - p - 1)
  BIC <- dev + p * log(n)

  # make a constant adjustment to AIC/AICc/BIC to give consistency with R's built
  # in AIC/BIC functions applied to glm objects
  # can do this because rxGlm() supplies AIC already (consistent with R/glm()) -
  # as long as computeAIC = TRUE in the function call

  deviance_constant_adjustment <- rx_glm$aic[1] - AIC

  AIC <- AIC + deviance_constant_adjustment
  AICc <- AICc + deviance_constant_adjustment
  BIC <- BIC + deviance_constant_adjustment

  cat("\n")
  cat("AIC: ", AIC, "\n")
  cat("AICc:", AICc, "\n")
  cat("BIC: ", BIC, "\n")
}
```

Let's test it...

```
data(mtcars)

glm_a1 <- glm(mpg ~ cyl + disp + hp + drat + wt + qsec + vs + am + gear + carb,
  data = mtcars,
  family = gaussian(link = "identity"),
  trace = TRUE)

glm_b1 <- rxGlm(mpg ~ cyl + disp + hp + drat + wt + qsec + vs + am + gear + carb,
```

```
data = mtcars,
family = gaussian(link = "identity"),
verbose = 1,
computeAIC = TRUE)

AIC(glm_a1)
AICc(glm_a1)
BIC(glm_a1)

wrc_information_criteria(glm_b1) # gives same results for glm_b1 as I got for
glm_a1
```

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answered Aug 23, 2020 at 9:55



Alan

671 6 24

in the [extractAIC manual page](#)

6



$$AIC(k) = -2\log(L) + k \times edf$$

Where :

- L is the likelihood and edf the equivalent degrees of freedom (i.e., the number of parameters for usual parametric models) of fit.
- For generalized linear models (i.e., for lm, aov, and glm), $-2\log L$ is the deviance, as computed by `deviance(fit)`.
- $k = 2$ corresponds to the traditional AIC, using $k = \log(n)$ provides the BIC (Bayes IC) instead.

Thus

$$BIC = AIC(k = \log(n)) = -2\log(L) + \log(n) \times edf$$

Edits following discussion in the comments and input of @user20650

- `glm_a1$rank` returns the number of fitted parameter without accounting for the fitted variance used in gaussian families.
- `?glm` states

deviance: up to a constant, minus twice the maximized log-likelihood. Where sensible, the constant is chosen so that a saturated model has deviance zero.

that's why `-2*logLik(glm_a1) - deviance(glm_a1) = 7.78 > 0`

- `summary(glm_a1)` returns the following line Dispersion parameter for gaussian family taken to be 7.023544 approximately the difference between $-2 \log$ likelihood and the deviance.

```
library(AICcmodavg)
#> Warning: package 'AICcmodavg' was built under R version 3.6.2
#> Warning: no function found corresponding to methods exports from 'raster' for:
#> 'wkt'
```

```

data(mtcars)

glm_a1 <- glm(mpg ~ cyl + disp + hp + drat + wt + qsec + vs + am + gear + carb,
             data = mtcars,
             family = gaussian(link = "identity"),
             trace = TRUE)
#> Deviance = 147.4944 Iterations = 1
#> Deviance = 147.4944 Iterations = 2

(loglik <- logLik(glm_a1))
#> 'log Lik.' -69.85491 (df=12)

# thus the degrees of freedom r uses are 12 instead of 11

n <- attributes(loglik)$nobs # following user20650 recommendation
p <- attributes(loglik)$df # following user20650 recommendation
dev <- -2*as.numeric(loglik)
my_AIC <- dev + 2 * p
my_AICc <- my_AIC + (2 * p^2 + 2 * p)/(n - p - 1)
my_BIC <- dev + p * log(n)

BIC(glm_a1)
#> [1] 181.2986
my_BIC
#> [1] 181.2986
AIC(glm_a1)
#> [1] 163.7098
my_AIC
#> [1] 163.7098
AICc(glm_a1)
#> [1] 180.1309
my_AICc
#> [1] 180.1309

```

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edited Aug 22, 2020 at 20:20

answered Aug 22, 2020 at 19:43



Abdessabour Mtk

3,888 2 14 21

-
- 3 Note that you can get `p` and `n` from `attributes(logLik(glm_a1))` . I think some of the confusion in op is that `deviance(glm_a1) != -2*logLik(glm_a1)` ; so looks like it is some constant out. – [user20650](#) Aug 22, 2020 at 19:48
-
- 2 @Alan; re AIC for Gaussian model. The extra parameter is for the variance term. Note that [some software](#) wont estimate this as usually we are interested in the difference in AIC and not as an absolute value . – [user20650](#) Aug 22, 2020 at 19:57
-
- 2 @Alan' ?`glm` states "deviance: up to a constant, minus twice the maximized log-likelihood. Where sensible, the constant is chosen so that a saturated model has deviance zero." if that helps ;/ – [user20650](#) Aug 22, 2020 at 20:02
-
- 1 the literal definition of degrees of freedom in statistics is the number of estimated parameters as [@user20650](#) stated we have 11 parameters for the vars and the intercept and one extra for the variance term – [Abdessabour Mtk](#) Aug 22, 2020 at 20:04
-
- 1 @Alan in the `summary(glm_a1)` it outputs the following Dispersion parameter for gaussian family taken to be 7.023544 which is approximately equal to the `-2logLik(glm_a1)-deviance(glm_a1)` – [Abdessabour Mtk](#) Aug 22, 2020 at 20:06
-